

An Analysis into the Prediction of Obesity Levels from Diet and Physical Activity Determinants in Latin American Adults

Abstract

This research analyzes important lifestyle factors associated with obesity by examining survey data from a Latin American demographic. The research aims to determine the relationship between physical activity, dietary habits, and obesity, while estimating the prevalence of obesity in the population, and verify the authenticity of the data in the dataset. It was found that a significant inverse relationship exists between physical activity and the weight categories where individuals categorized as underweight and normal weight appear to have similar frequencies of physical activity compared to individuals that are categorized as overweight and obese. In contrast, the number of main meals consumed daily shows it is a negligible determinant with body weight. Note that within the sample, the estimated population proportion of overweight and obese individuals is between 71.6% and 75.4% which implies every 3 in 4 individuals are overweight or obese. These results indicate that within this demographic, physical activity is a significant determinant that can help distinguish weight categories, however cannot be said the same for the frequency of meals. Consequently, the high proportion of overweight and obese individuals highlight a substantial public health concern that requires prioritizing not just increasing the frequency of physical activity due to limitations on the data of the dataset.

Introduction

Obesity is a widespread medical condition and is a serious global health concern. This report analyzes survey data collected from a 2023 dataset of survey responses of individuals residing in Latin America, including Mexico, Peru, and Colombia to determine primary lifestyle factors of obesity. By seeking to identify predictive patterns in dietary habits and physical activity, it provide insights into the underlying causes of this health hazard within this Latin American demographic.

This report aims to answer the following questions for the dataset:

1. Do individuals with higher obesity levels report lower average frequencies of physical activity?
2. What is the proportion of adults who are classified as overweight or obese, and can we estimate this proportion with a 95% confidence interval, assuming the data is from a simple random sample without replacement?
3. What is the relationship between the number of main meals consumed per day and body weight among adults in the dataset?
4. Do obesity categories differ in mean body weight?

Variable Information

Table 1: Mini codebook for all variables used in the report.

Variable Name	Description	Type	Original Values	Recoding (if applicable)
NObeyesdad	Classification of obesity based on BMI	Categorical (nominal)	Underweight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, Obesity_Type_III	"Insufficient_Weight": (Underweight), "Overweight": (Overweight_Level_I, Overweight_Level_II), "Obese": (Obesity_Type_I, Obesity_Type_II, Obesity_Type_III)
FAF	Frequency of physical activity	Numerical (ordinal)	0, 1, 2, 3	—
NCP	Number of major meals consumed per day	Numerical (integer)	Positive values only	—
Weight	Weight of individuals (kilograms)	Numerical (continuous)	Positive values only	—

All variables were analyzed according to their stated type and values unless otherwise noted. The dataset contained no missing values for the selected variables.

Results

This section presents the findings from the statistical analysis of the dataset, which employs ANOVA, confidence interval estimation, and correlation analysis to address the four research questions, and reveals several key patterns that provide significant inferences into the lifestyle factors associated with obesity within this demographic.

Question 1: Do individuals with higher obesity levels report lower average frequencies of physical activity?

This section seeks to determine whether average physical activity frequency (FAF) differs across the obesity classes. Below, summary statistics in Table 2 and an ANOVA table in Table 3 were compute, and visualized using a boxplot (Fig. 1) to compare and contrast the distribution of physical activity across the four body weight categories.

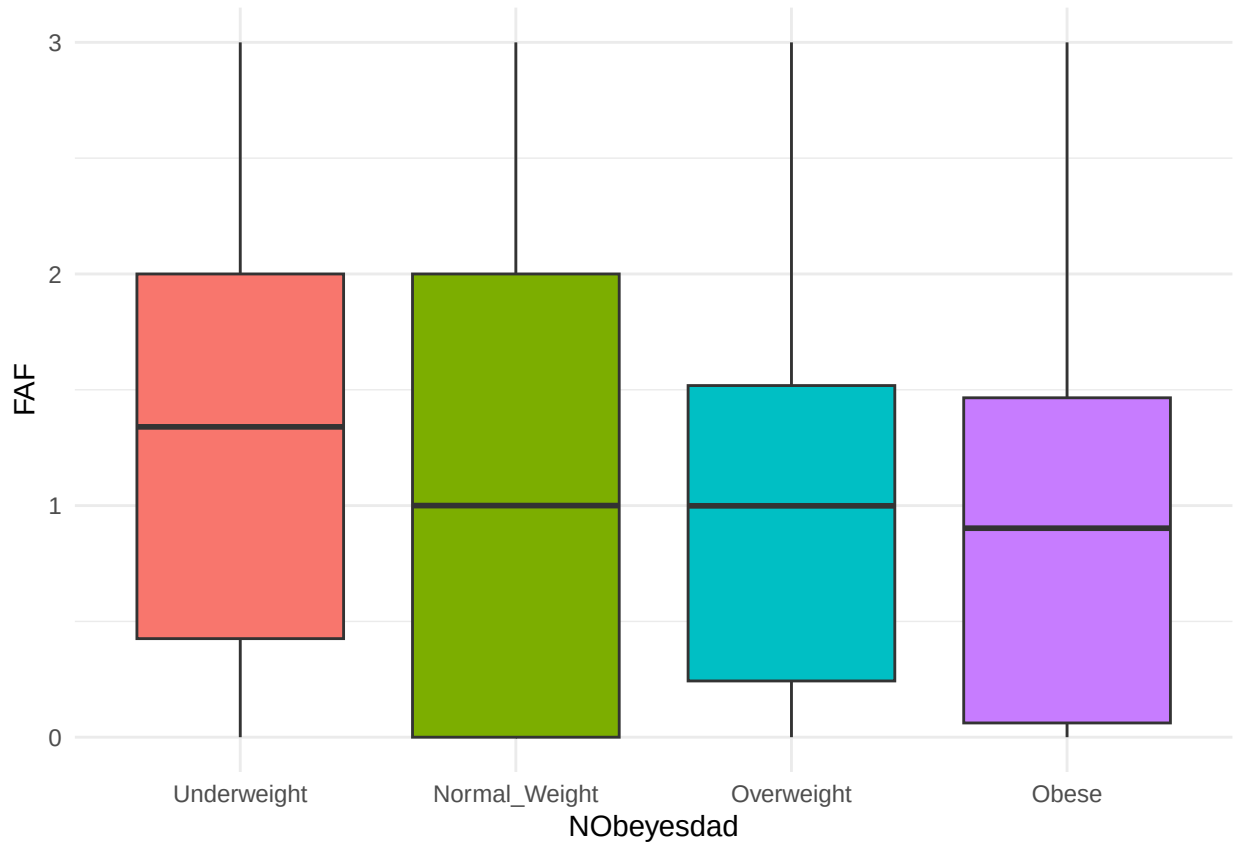


Figure 1: Relationship between classes of obesity and physical activity frequency

Table 2: Summary statistics of physical activity frequency (FAF) across classes of obesity

Obesity Classification	N	Mean	SD	Median	IQR
Underweight	272	1.25	0.86	1.34	1.57
Normal_Weight	287	1.25	1.02	1.00	2.00
Overweight	580	1.01	0.84	1.00	1.28
Obese	972	0.87	0.77	0.90	1.40

Observe the patterns seen in Figure 1 and Table 2 which show individuals classified as “Obese” report the

lowest median physical activity frequency at 0.9 per week. By contrast, individuals classified as “Underweight” report the highest median physical activity frequency at 1.34 per week. As a result, this indication reveals a generally inverse relationship between body weight and the amount of physical activity received each week. Also note that in Figure 1, the interquartile ranges of adjacent weight classes substantially overlap. That is, the middle 50% around the median of each weight class are roughly relative to the next weight category, indicating that an individual’s weight cannot be determined by the frequency of physical activity alone.

Table 3: One-way ANOVA results for physical activity frequency across obesity classifications

Term	Df	Sum.Sq	Mean.Sq	F.value	Pr..F.
Obesity Classification	3	49.61	16.54	23.59	5.15e-15
Residuals	2107	1477.00	0.70		

Table 4: Pairwise comparisons between obesity classifications

Comparison	Difference	Lower CI	Upper CI	p-value
Normal_Weight-Underweight	-0.003	-0.185	0.179	1.000
Overweight-Underweight	-0.243	-0.401	-0.085	0.000
Obese-Underweight	-0.375	-0.523	-0.228	0.000
Overweight-Normal_Weight	-0.240	-0.395	-0.085	0.000
Obese-Normal_Weight	-0.372	-0.517	-0.228	0.000
Obese-Overweight	-0.133	-0.245	-0.020	0.014

To determine if the different frequencies of physical activity per week were statistically significant, a one-way ANOVA was conducted. In Table 3, the p-value of the test was $5.15 \times 10^{-15} < 0.001$, which provides very strong evidence that at least one obesity class has a differing mean in physical activity.

Then, to narrow down which specific categories of weight significantly differ, Table 4 performs pairwise differences between each of the different weight classes. Notably, this analysis reveals a distinct pattern. Between individuals in “Underweight” and those in “Normal_Weight”, the difference in physical activity is negligible and not statistically significant (p-value > 0.001) which indicates individuals within these two categories maintain similar levels of physical activity. However, compared to individuals in the “Overweight” and “Obese” categories, a significant reduction in physical activity can be observed.

Overall, the analysis provides an answer that clearly shows individuals in different weight categories have differing averages in their frequency of physical activity. This relationship is generally inverse and shows a critical threshold whereby differences between individuals in the “Underweight” and “Normal_weight” categories are not as profound when comparing to individuals classified as “Overweight” or “Obese”. Therefore, lower levels of physical activity leads to strong association with being overweight or obese.

Question 2: What is the proportion of adults who are classified as overweight or obese, and can we estimate this proportion with a 95% confidence interval, assuming the data is from a simple random sample without replacement?

This section aims to quantify the proportion of overweight and obese individuals within the sample in order to provide a rough estimate of the obesity problem for the broader population. Below, summary statistics in Table 5 and a Chi-squared proportion test in Table 6 were compute, and visualized using a bar graph (Fig. 2) to contrast the distribution of underweight and normal weight individuals versus individuals with excessive weight.

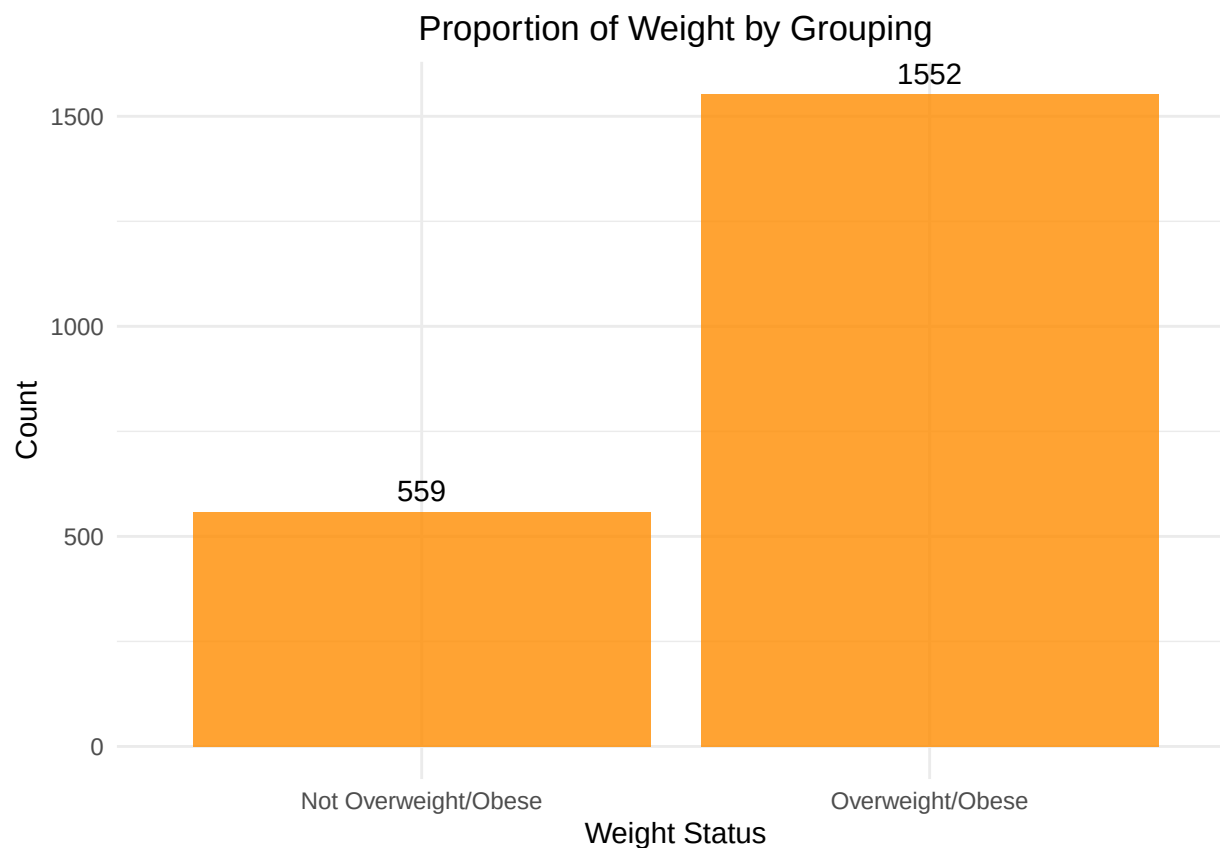


Figure 2: Proportion of individuals by weight

Table 5: Proportion of Weight by Grouping

Weight Groups	Count	Proportion	Percentage
Not Overweight/Obese	559	0.265	26.5
Overweight/Obese	1552	0.735	73.5

Figure 2 and Table 5 provide information showing that 73.5% individuals out of 2111 individuals were classified as overweight or obese, while the remaining 26.5% of 2111 individuals were classified as not overweight or obese.

A one-sample Chi-squared test was conducted to help mainly confirm that the sample provides sufficient evidence to make inferences on the population. Since the p-value is $2 \times 10^{-16} < 0.001$, which is really small, the test confirms the observed proportion indeed provides strong evidence for making population inferences.

Table 6: Results of proportion test for the overweight/obese proportion

	Test	Successes	Trials	Sample Proportion	Test Statistic	P-value	95% CI Lower	95% CI Upper
X-squared	One-sample proportion test	1552	2111	0.735	467.1	<2e-16	0.716	0.754

The analysis reveals that approximately 73.5% of individuals within the sample are affected by obesity. Based on the 95% confidence interval conducted from the Chi-squared test, we can be 95% confident that the true population proportion of individuals that are overweight or obese falls between 71.6% and 75.4%. Hence, obesity is a serious health concern that affects approximately 3 in 4 individuals in this Latin American demographic.

Question 3: What is the relationship between the number of main meals consumed per day and body weight among adults in the dataset?

This section aims to investigate whether the relationship between the number of main meals consumed daily (NCP) and body weight have any correlation. Below, summary statistics in Table 7, correlation in Table 8, linear regression in Table 9 and 10, were computed and visualized with a scatterplot (see Fig. 3) to quantify the relationship.

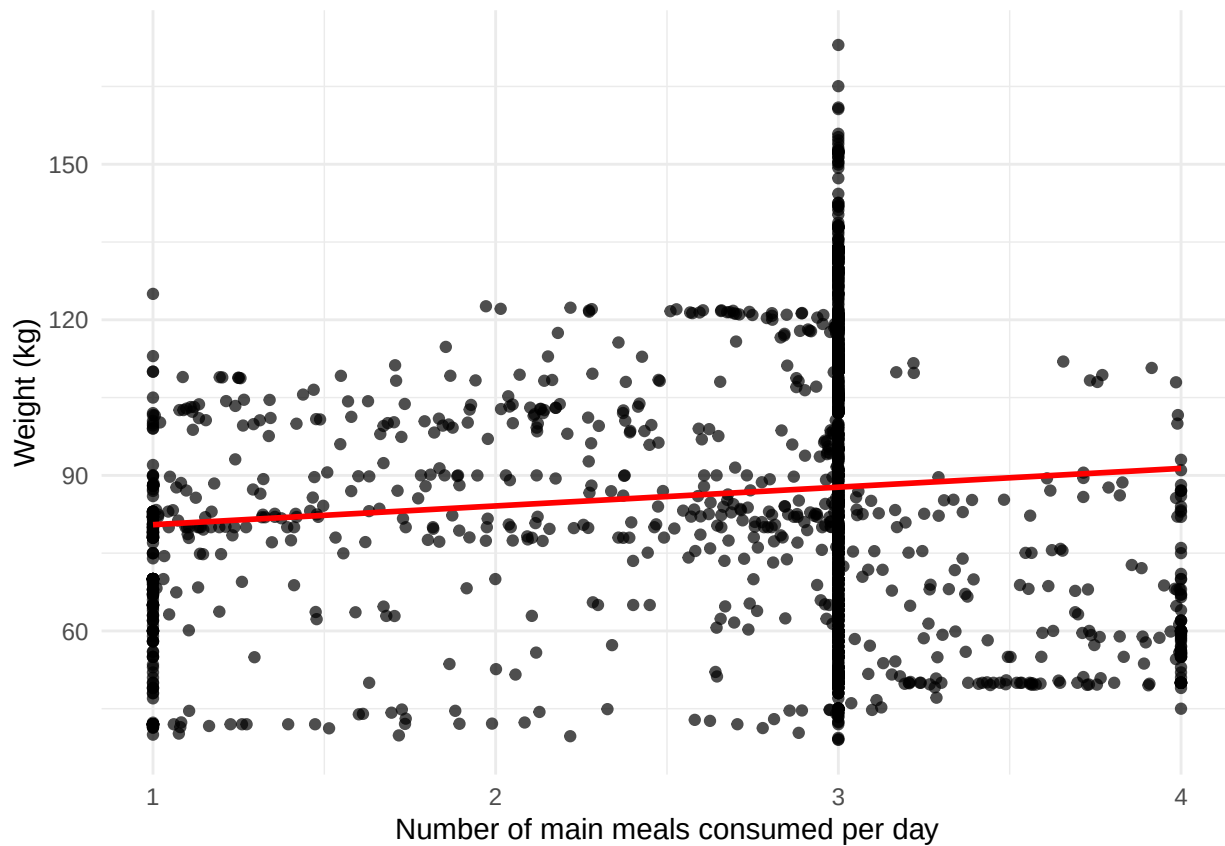


Figure 3: Relationship between number of main meals and weight

Table 7: Summary statistics for Weight (kg)

Statistic	Value
Mean	86.59
Standard Deviation	26.19
Minimum	39.00
Maximum	173.00
Median	83.00
IQR	41.96

Table 8: Correlation between meals per day and weight

	Correlation	Test Statistic	P-value	95% CI Lower	95% CI Upper
t	0.107	4.964	7.45e-07	0.065	0.149

Figure 3 displays a positive trend which suggests that more main meals consumed per day is associated with higher body weight. Table 8 further supports this analysis with a small p-value $7.45 \times 10^{-7} < 0.001$. Therefore, there is strong evidence to suggest that the correlation is positive and there exists a positive relationship between number of main meals per day and body weight. Additionally, this analysis can be further corroborated by the 95% confidence interval of the correlation test that is completely positive, indicating it is unlikely that the true relationship is zero or negative.

Table 9: Linear regression between Weight and Main Meals per Day (NCP)

	Term	Estimate	Std. Error	t-value	P-value
(Intercept)	Intercept	76.870	2.038	37.725	< 2e-16
NCP	NCP	3.618	0.729	4.964	7.45e-07

Table 10: Regression summary

Statistic	Value
R-squared	0.012
Adj. R-squared	0.011
F-statistic	24.6
P-value	7.45e-07
Residual SE	26

Note that the linear regression model produces a similar outcome. It predicts for each additional main meal consumed per day, body weight is to increase by 3.618 kg. However, understand that the R-squared value indicates that this prediction only explains about 1.2% of the total variation in body weight. Thus, the number of daily main meals cannot be practically significance enough to deduce an individual's body weight.

Although, there exists a statistically significant linear relationship between the number of main meals per day and body weight, the relationship is not strong. This relationship does have a correlation that results in determining body weight, but nevertheless, cannot be a strong and meaningful predictor of an individual's body weight in the population.

Question 4: Do obesity categories differ in mean body weight?

This section investigates whether obesity categories have distinct ranges of body weight. Below, summary statistics in Table 11, a one-sided ANOVA test in Table 12, and pairwise comparisons in Table 14, are computed and visualized with a boxplot (see Fig. 4) to visually compare and contrast the range of body weights for each category.

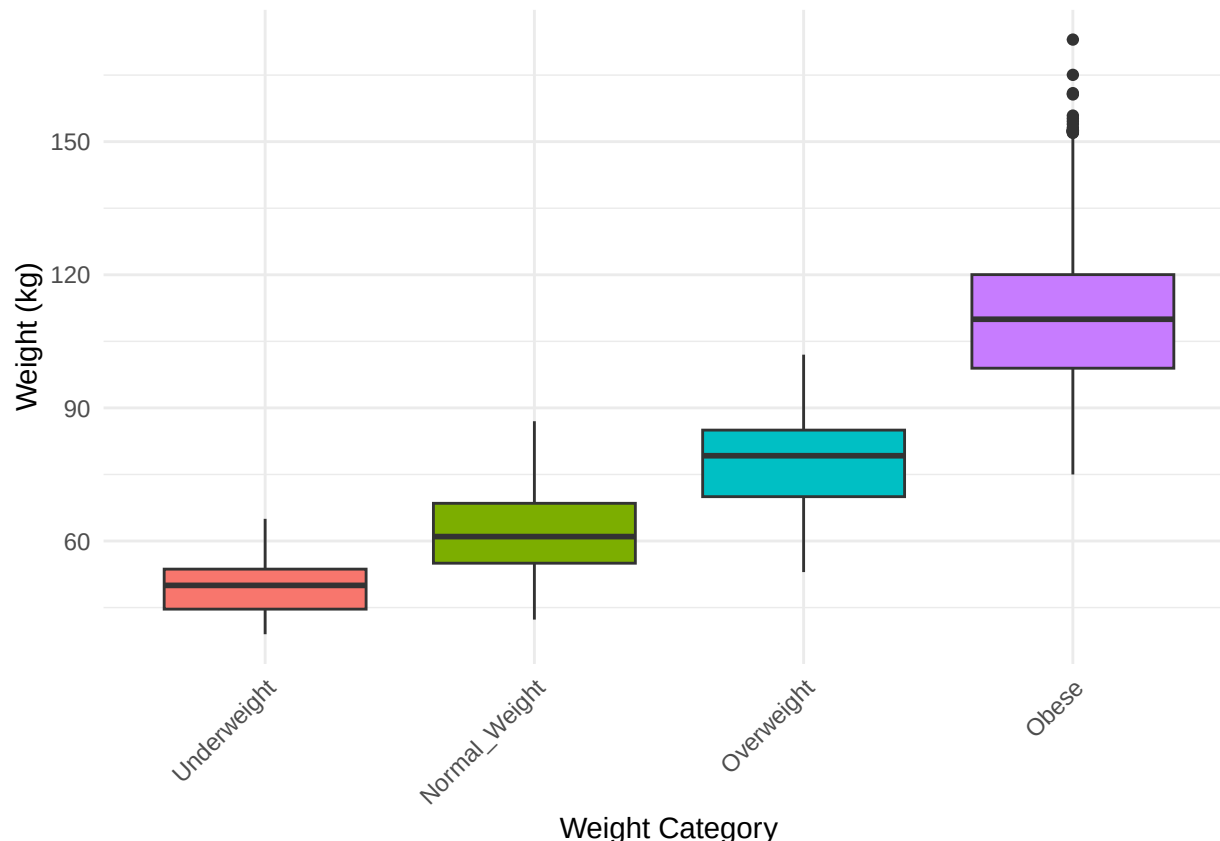


Figure 4: Weight distribution across obesity categories

Table 11: Summary statistics across obesity categories

Weight Category	N	Mean	SD	Median	IQR	Min	Max
Underweight	272	49.91	6.01	50.00	9.01	39.0	65
Normal_Weight	287	62.16	9.30	61.00	13.50	42.3	87
Overweight	580	78.18	9.32	79.22	15.00	53.0	102
Obese	972	109.08	17.36	109.98	21.09	75.0	173

Observe that Figure 4 shows visual confirmation of distinct weight categories that have separate distributions with minimal overlap. Furthermore, there is a clear, increasing trend in median weight as the weight categories become progressively heavier. Moreover, the summary statistics support this claim with increasing means; from 49.91 kg for “Underweight” individuals, to 109.08 kg for “Obese” individuals.

A one-way ANOVA test was conducted for Table 11. The result was that Table 12 shows a p-value = $2 \times 10^{-16} < 0.0001$ which is very significant and provides very strong evidence that not all weight categories share the same mean body weight. Table 13 further corroborates these findings by testing and comparing the differences between different weight categories. Note that for each test between two different weight categories, the p-value has been significant for all of them. As a result, understand that all these differences

Table 12: One-way ANOVA results for weight across obesity categories

Term	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Weight Category	3	1070187.2	356729.1	1992.52	<2e-16
Residuals	2107	377225.2	179.0		

Table 13: Pairwise comparisons of mean weight across obesity categories

	Difference	CI Lower	CI Upper	P-value
Normal Weight - Underweight	12.249	9.338	15.160	< 0.001
Obesity - Underweight	28.270	25.742	30.798	< 0.001
Overweight - Underweight	59.176	56.816	61.536	< 0.001
Obesity - Normal Weight	16.021	13.538	18.504	< 0.001
Overweight - Normal Weight	46.927	44.616	49.238	< 0.001
Overweight - Obesity	30.906	29.101	32.711	< 0.001

are significant and substantial enough to represent that the mean weight of all weight categories are not the same.

In summary, the analysis proves that all four obesity categories differ vastly in their mean body weights, with their differences being both statistically significant and large. Each category represents a distinct and separate range in the spectrum of weight classes which validate the use of these four categorical weight classifications within the broader population.

Conclusion

This report provides a comprehensive understanding of the relationships between lifestyle determinants and obesity in this Latin American demographic. The results show that physical activity is a key indicator for determining an individual's weight category while on the other hand, dietary patterns measured by meal frequency plays a negligible role in this regard.

Most notably, the analysis of research question 1 reveals a significant and unambiguous inverse relationship with obesity categories. It is observed that there exists a 'rift' where individuals in the categories underweight and normal weight appear statistically similar however, the inverse relationship appears much more prominently when observing individuals in the overweight and obese categories. Thus, this suggests that maintaining a normal weight does not require frequent physical activity, but failing to attain a certain frequency of physical activity strongly associates with being overweight or obese.

Also, the report discovers through research question 2 that the proportion of the sample affected by overweight or obesity is 73.5%. Nevertheless, with 95% confidence, the true population proportion of individuals that are obese should be between 71.6% and 75.4% which indicates that approximately 3 in 4 adults within this demographic faces substantial weight-related health risks.

By contrast, analyzing research question 3 shows that the relationship between the number of main meals versus body weight does show a slight linear relationship, although it is exceptionally weak in being a significant determinant of an individual's body weight.

Finally, research question 4 confirms that the categorization of body weight into underweight, normal weight, overweight, and obese, represent distinct categories that differ significantly from each other in mean body weight. As a result, this analysis confirms that the body weight categories capture meaningful information about individuals in each weight class.

While the analysis offers valuable insights into determining factors of obesity, it should be noted that several limitations in the dataset should be acknowledged. For example, the dataset derives data mainly from a survey sent out to individuals in the chosen Latin American demographic, which is highly susceptible to bias and interpretation based on individuality. This creates inaccuracies in the data, which is problematic when trying to accurately understand the extent of the problem of obesity in the population.

Another limitation is a lack of detail that the variable NCP reports. Research question 3 at most compares weight to the daily number of meals, but cannot distinguish the composition or caloric intake an individual consumes per meal. Thus, this leads to categorizing any individual eating large, high caloric meals together with those eating small, balanced meals, which severely alters the impact that dietary has on body weight.

Furthermore, the report does not factor in any other determinants except frequency of physical activity (FAF) and number of daily main meals. For example, an individual's ancestry, socioeconomic status, sleep patterns, or stress, could all be considerable determinants that influence body weight. As a result, missing such crucial information about an individual likely severely impacted the results of the analysis on the dataset.

Next Steps

To better reflect obesity in this Latin American demographic, several improvements could be made for collecting data. If surveys are still used in a future data collection session, the survey could be improved by added more metrics that better track the activities of an individual. In other words, the survey could ask about consumption of fruits, vegetables, or carbohydrates, as well as how much sugary beverages and fast food is consumed in a month. Further optimizations could include asking about a rough estimate on weight of food consumed daily or caloric intake daily. When asking about physical activity, the survey should detail specific sub-questions on types of physical activity and how long these physical activities last to avoid as much subjectivity as possible. Additionally, new determinants that are highly prevalent in determining an individual's body weight should be included in the survey. For example, an individual's socioeconomic

status could highly reflect how much free time they have which directly correlates to the level of stress of an individual and can also determine whether they have access to determinants of physical activity.

References

Mehrpavar, F. (2023). *Estimation of obesity levels in individuals from the countries of Mexico, Peru and Colombia* [Dataset]. Latin America. Retrieved from <https://www.kaggle.com/datasets/fatemehmehrpavar/obesity-levels>

Appendix

Research Question 1

```
library(ggplot2)
data <- read.csv("ObesityDataSet_raw_and_data_sinthetic.csv")
data$NObeyesdad[data$NObeyesdad %in%
                 c("Obesity_Type_I", "Obesity_Type_II", "Obesity_Type_III")] <- "Obesity"
data$NObeyesdad[data$NObeyesdad %in%
                 c("Overweight_Level_I", "Overweight_Level_II")] <- "Overweight"
myaov <- aov(FAF~NObeyesdad, data)
summary(myaov)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## NObeyesdad      3   49.6   16.535    23.59 5.15e-15 ***
## Residuals    2107 1477.0    0.701
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(myaov)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = FAF ~ NObeyesdad, data = data)
##
## $NObeyesdad
##              diff              lwr              upr
## Normal_Weight-Insufficient_Weight -0.002744575 -0.18490494  0.17941579
## Obesity-Insufficient_Weight        -0.375243562 -0.52290463 -0.22758250
## Overweight-Insufficient_Weight      -0.242697200 -0.40089292 -0.08450148
## Obesity-Normal_Weight              -0.372498987 -0.51711360 -0.22788438
## Overweight-Normal_Weight           -0.239952625 -0.39530861 -0.08459664
## Overweight-Obesity                 0.132546362  0.01960003  0.24549269
##              p adj
## Normal_Weight-Insufficient_Weight 0.9999791
## Obesity-Insufficient_Weight        0.0000000
## Overweight-Insufficient_Weight      0.0004791
## Obesity-Normal_Weight              0.0000000
## Overweight-Normal_Weight           0.0004294
## Overweight-Obesity                 0.0137458
```

Research Question 2

```
x <- sum(data$NObeyesdad %in% c("Obesity", "Overweight"))
n <- nrow(data)
prop.test(x, n, correct = F)
```

```
##
## 1-sample proportions test without continuity correction
##
## data:  x out of n, null probability 0.5
## X-squared = 467.1, df = 1, p-value < 2.2e-16
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
```

```
## 0.7159595 0.7535792
## sample estimates:
##      p
## 0.7351966

data$OverweightObese <- ifelse(
  data$NObeyesdad %in% c("Overweight", "Obesity"),
  "Overweight/Obese", "Not Overweight/Obese"
)

counts <- table(
  WeightStatus = ifelse(data$NObeyesdad %in% c("Obesity", "Overweight"),
    "Overweight_or_Obese", "Other")
)
counts

## WeightStatus
##           Other Overweight_or_Obese
##           559          1552

round(prop.table(table(data$NObeyesdad)), 3)

##
## Insufficient_Weight      Normal_Weight      Obesity      Overweight
##           0.129           0.136           0.460           0.275
```

Research Question 3

```
model <- lm(Weight~NCP, data)
summary(model)

##
## Call:
## lm(formula = Weight ~ NCP, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -48.723 -20.536  -2.488   21.320   85.277
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  76.8702     2.0377   37.725 < 2e-16 ***
## NCP          3.6177      0.7288    4.964 7.45e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 26.05 on 2109 degrees of freedom
## Multiple R-squared:  0.01155,    Adjusted R-squared:  0.01108
## F-statistic: 24.64 on 1 and 2109 DF,  p-value: 7.454e-07

par(mfrow=c(2,2))
```

Research Question 4

```

myaov2 <- aov(Weight~NObeyesdad, data)
summary(myaov2)

##              Df Sum Sq Mean Sq F value Pr(>F)
## NObeyesdad    3 1070187  356729    1993 <2e-16 ***
## Residuals   2107  377225    179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

TukeyHSD(myaov2)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Weight ~ NObeyesdad, data = data)
##
## $NObeyesdad
##              diff              lwr              upr p adj
## Normal_Weight-Insufficient_Weight 12.24872    9.337571   15.15987    0
## Obesity-Insufficient_Weight      59.17601   56.816205   61.53582    0
## Overweight-Insufficient_Weight    28.26972   25.741554   30.79788    0
## Obesity-Normal_Weight            46.92729   44.616169   49.23841    0
## Overweight-Normal_Weight          16.02100   13.538214   18.50378    0
## Overweight-Obesity               -30.90629  -32.711318  -29.10127    0

category_counts <- as.data.frame(table(data$NObeyesdad))
category_counts$Proportion <- round(prop.table(category_counts$Freq), 3)
category_counts$Percentage <- round(prop.table(category_counts$Freq) * 100, 1)

colnames(category_counts) <- c("Weight Category", "Frequency", "Proportion", "Percentage")

kable(category_counts,
      format = "latex",
      booktabs = TRUE,
      caption = "Proportions of obesity categories") %>%
  kable_styling(
    latex_options = c("hold_position"),
    font_size = 9
  ) %>%
  row_spec(0, bold = TRUE)

```

Table 14: Proportions of obesity categories

Weight Category	Frequency	Proportion	Percentage
Insufficient_Weight	272	0.129	12.9
Normal_Weight	287	0.136	13.6
Obesity	972	0.460	46.0
Overweight	580	0.275	27.5